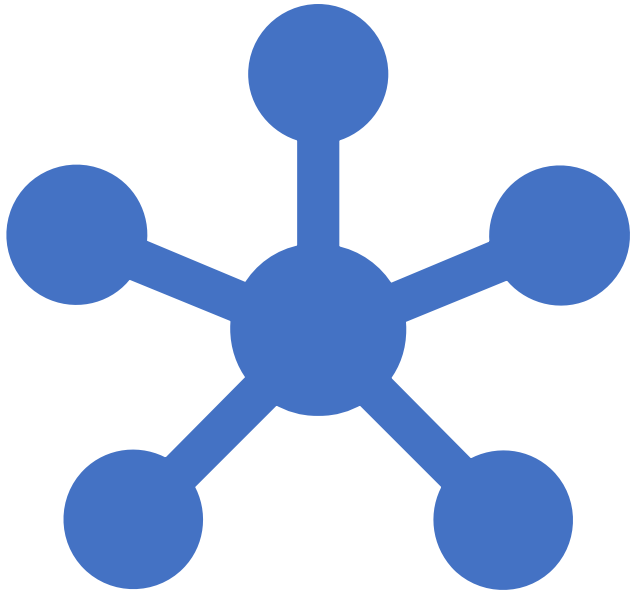


Relay Logic Ladder Diagram - 2

More about Relay Logic

Objectives

- Course Learning Objective #1: Students should recognize and understand the Relay Logic Programming and be able to compare it later with the Programmable Logic Control (PLC) of following modules.
- Specifically, at the end of this class, students should:
  - MLO #1.8: Use symbols to represent different types of input and output devices.
  - MLO #1.9: Create relay logic diagrams using the standard relay logic rules.
  - MLO #1.10: Create relay logic circuits for process and industrial control problems.
  - MLO #1.11: Distinguish between Relay Logic (Ladder) Diagrams (RLDs) and Ladder Diagrams ((PLC) ladder diagrams)

Relay Logic Diagrams (1)

- Sometimes called *elementary diagrams*, *line diagrams*, or *relay ladder logic (RLL)*
- From here on out, we will use the term *relay logic diagram* (RLD) to describe the relay ladder logic diagram
- It may help to create a PLC ladder diagram. PLC ladder diagrams will be considered later.
- RLDs show symbols for input and output devices. PLC ladder diagrams don’t

Relay Logic Diagrams (2)

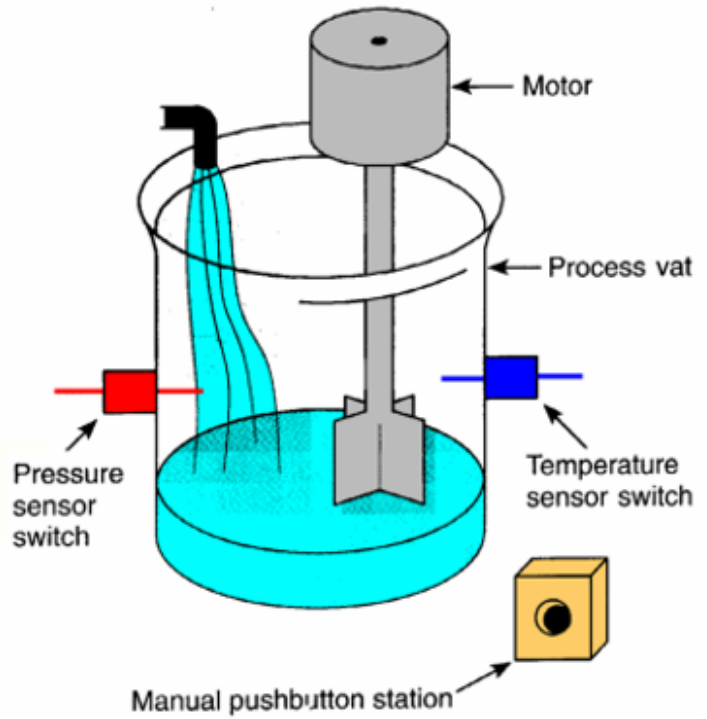
- After the relay logic diagram is drawn, it can be converted to a PLC ladder logic diagram, often called a *ladder diagram*.
- However, we don’t need an RLD as a first step to draw a PLC ladder diagram. We will draw PLC ladder diagrams directly.
- A Ladder diagram:
  - Is used to program the PLC
  - Defines the operations that the PLC must perform
  - More about PLC diagrams later

RLD – Example 1 (1)

PLC Mixer Process Control Problem

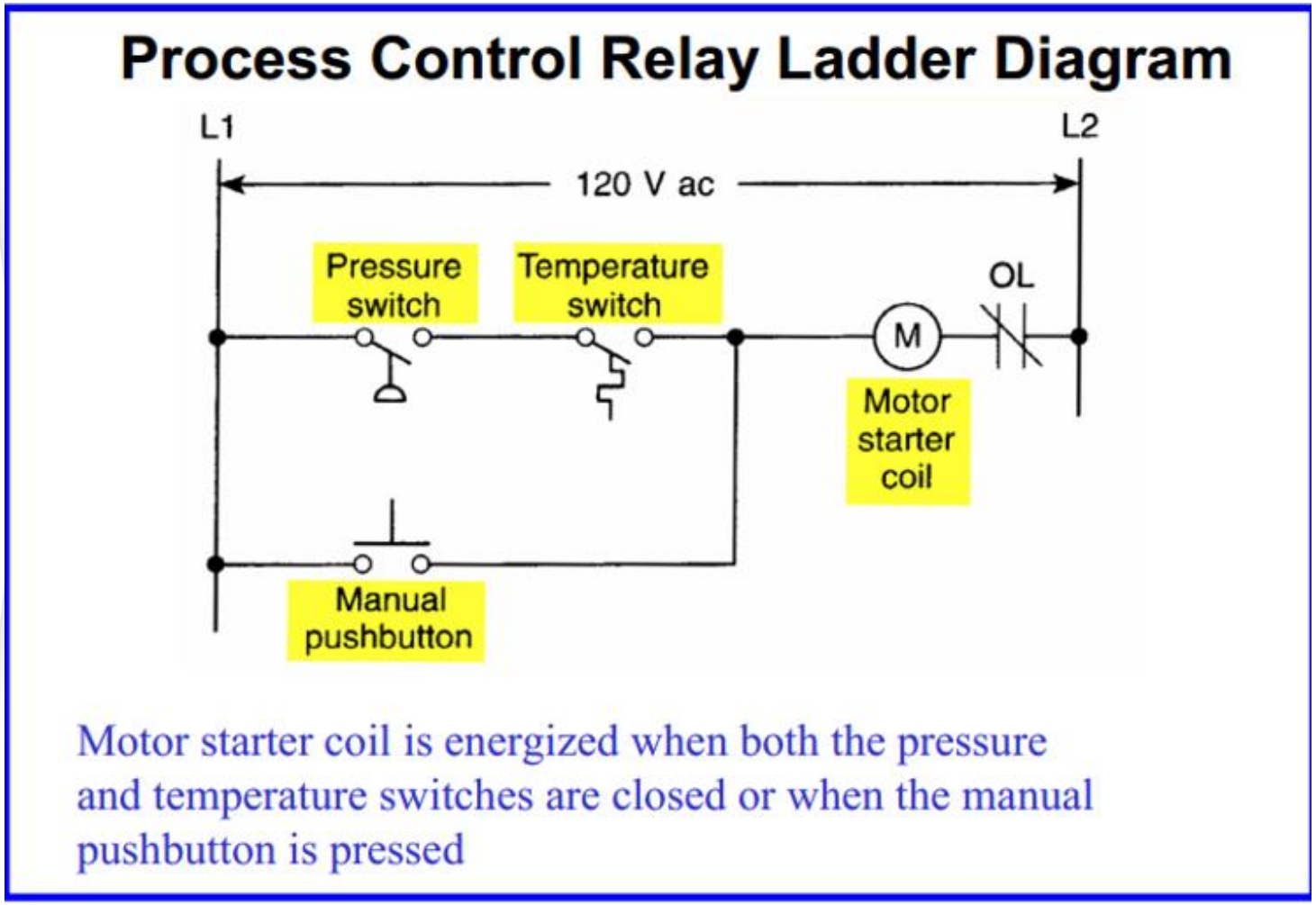
Mixer motor to automatically stir the liquid in the vat when the temperature and pressure reach preset values

Alternate manual pushbutton control of the motor to be provided (Jog button, NO)



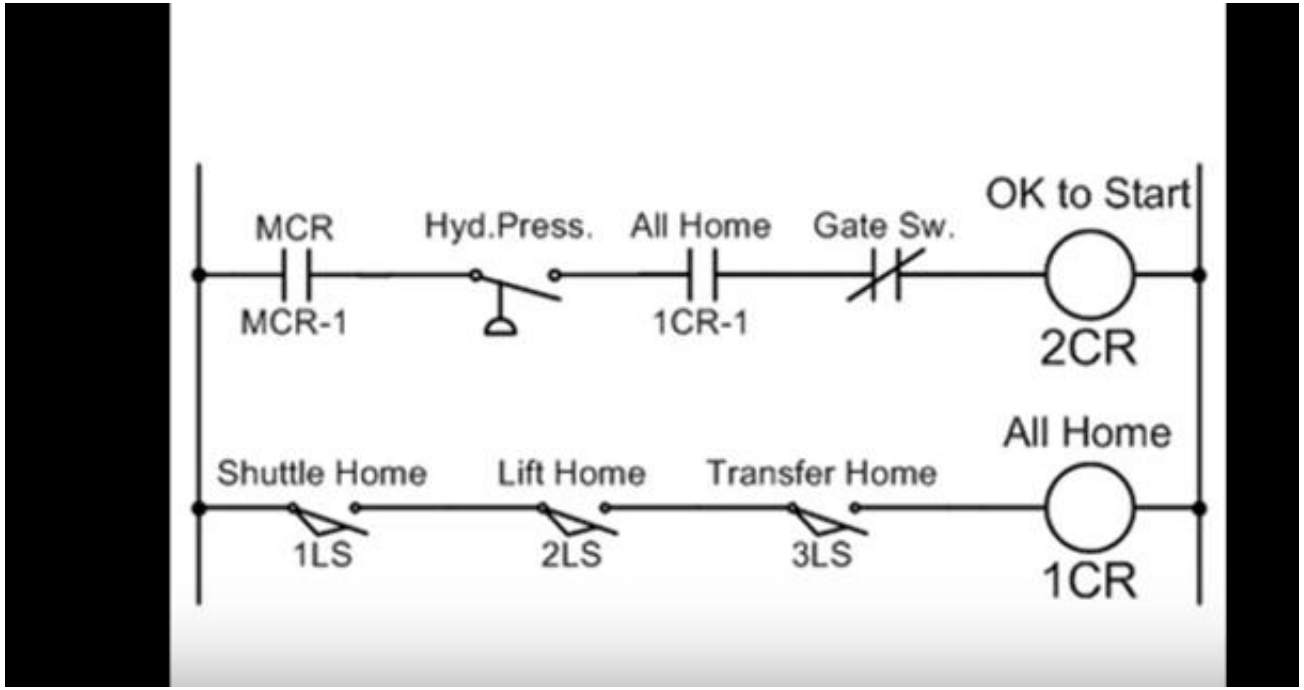
The temperature and pressure sensor switches close their respective contacts when conditions reach their preset values

RLD – Example 1 (2)



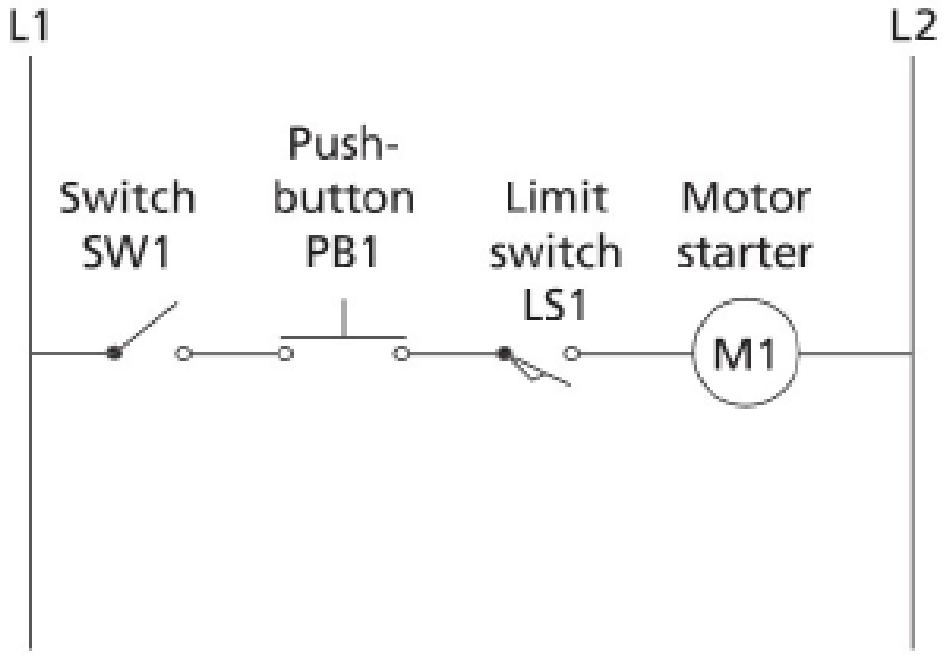
RLD – Example 2

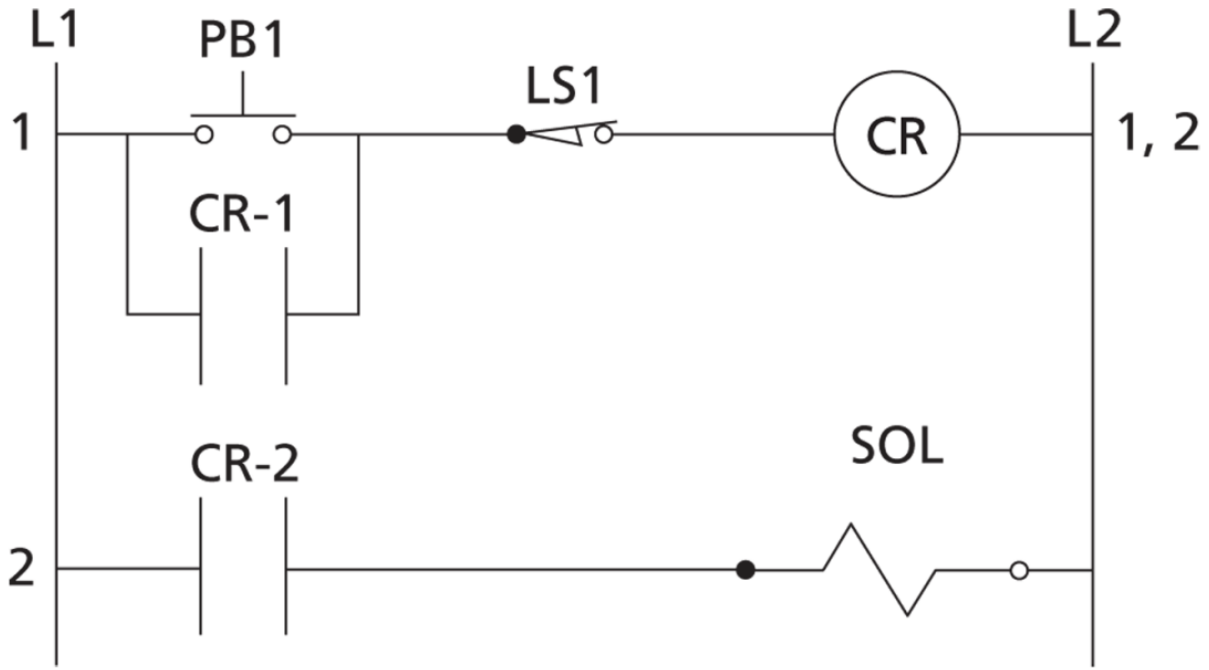
- Describe the operation of the RLD shown next (when is it OK to start?):



RLD – Example 3

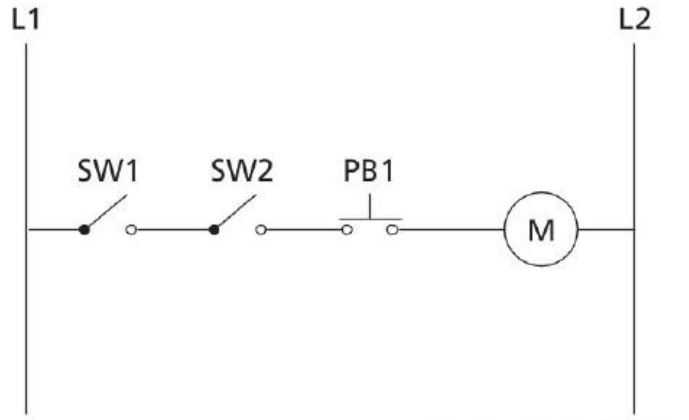
When the motor 1 is on? When the solenoid is on?





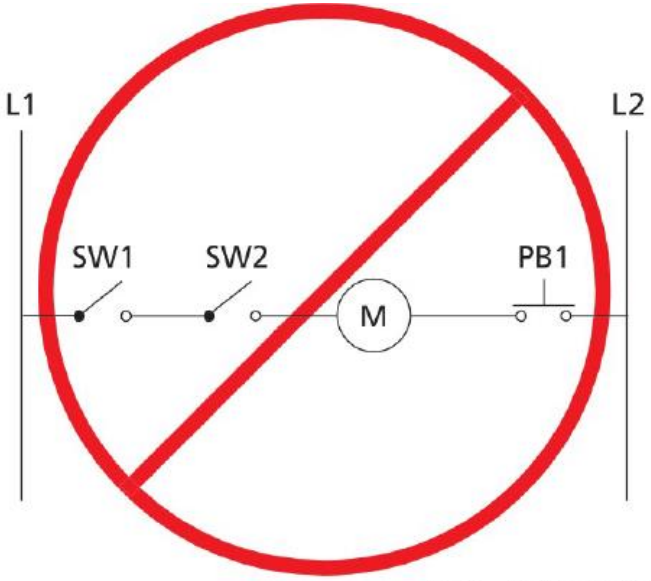
Relay Logic Diagrams Rules (1)

- **Rule 1.** Input devices are placed near the left corner of the rung (Figs. 5-6 and 5-7)
- **Rule 2.** One (and only one) output is placed near the right corner of the rung (Figs. 5-6 and 5-7)



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Figure 5-6. Correct relay logic diagram. The input devices are placed on the left side and the output device is on the far right.



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Figure 5-7. An incorrect relay logic diagram. The motor, which is an output device, must be placed on the right side.

Relay Logic Diagrams Rules (2)

- **Rule 3.** Input devices can be connected in series, parallel, or combination (Fig. 5-8)
- **Rule 4.** Output devices *cannot* be connected in series (Figs. 5.9, and 5.10 in next slide), but they can be connected in parallel

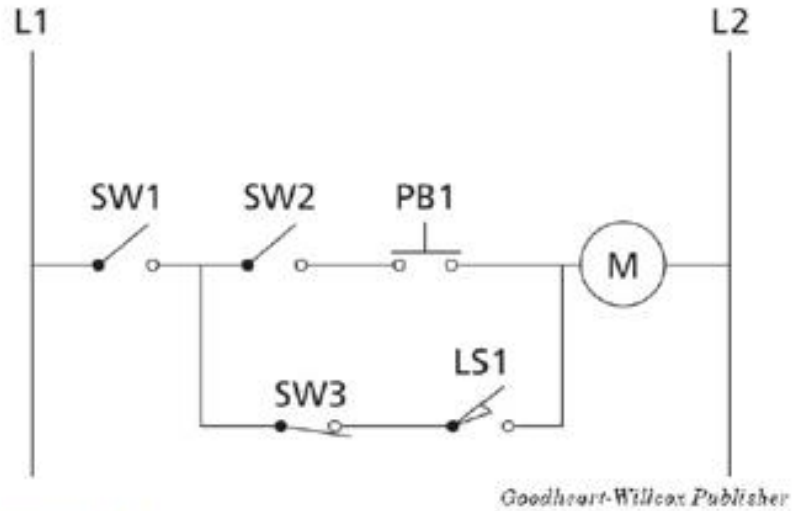


Figure 5-8. Combination of series and parallel input devices in a rung.

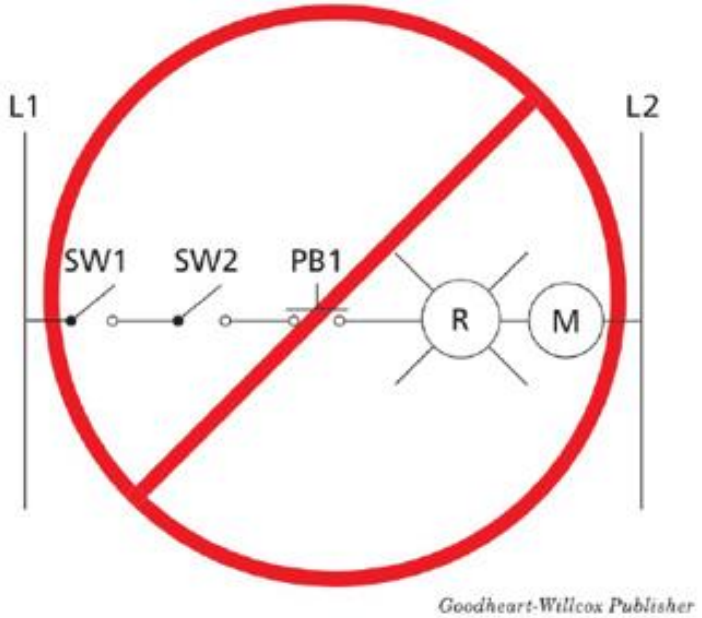


Figure 5-9. An incorrect relay logic diagram. Output devices cannot be placed in series.

Relay Logic Diagrams Rules (3)

- **Rule 5:** Input devices that need to be represented more than once are made so by using relays with multiple poles. An output device can only be represented once in the diagram. (Figs. 5-10, 5-11 and 5-11a)

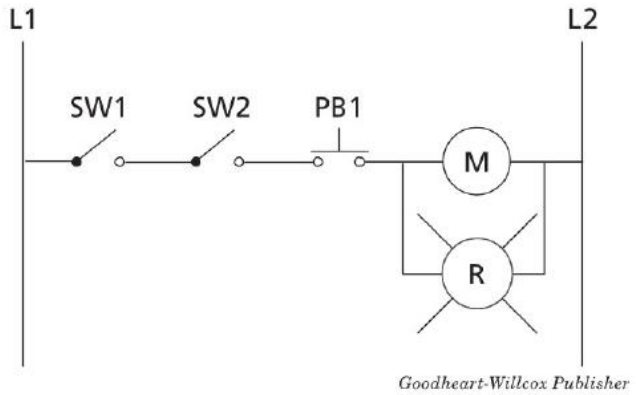


Figure 5-10. A correct relay logic diagram in which output devices are in parallel.

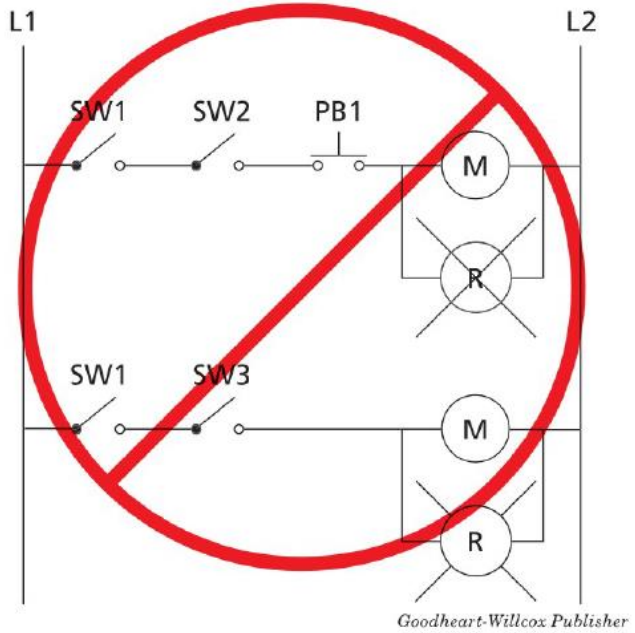


Figure 5-11. An incorrect relay logic diagram. Each output symbol can be used only once. and SW1 is used twice

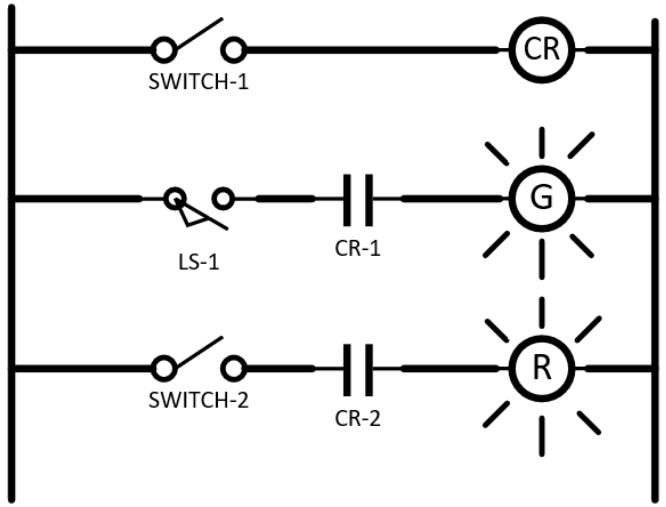


Fig. 5-11a. Correct RLD. Switch-1 is used twice through a relay with at least two NO poles

Relay Logic Diagrams Rules (4)

- **Rule 6.** All the input and output devices must be placed horizontally

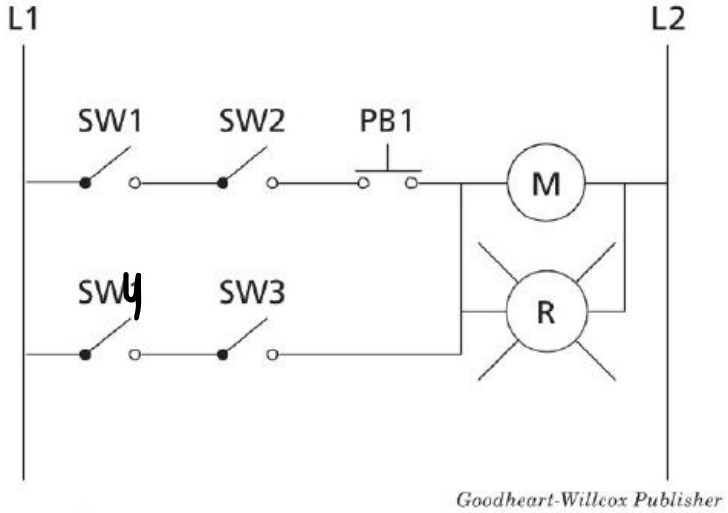


Figure 5-12. A correct relay logic diagram. Only one output device symbol is used for each output displayed.

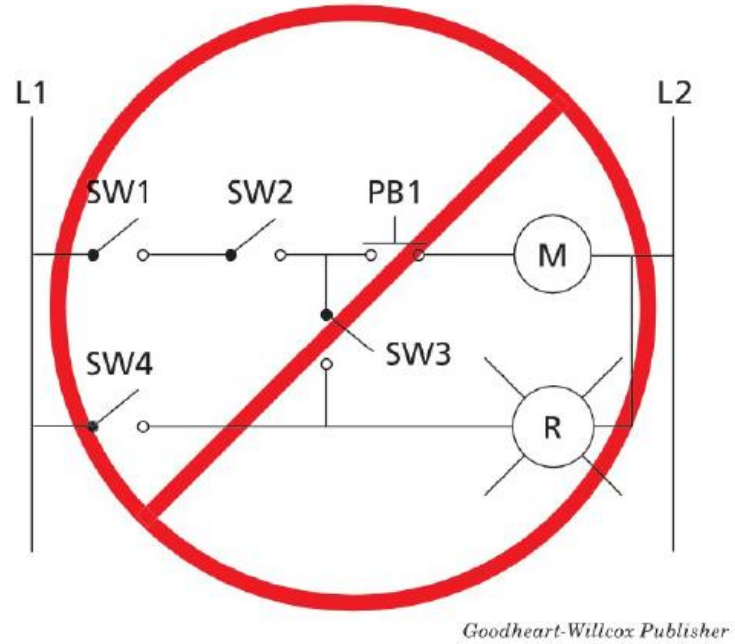


Figure 5-13. An incorrect relay logic diagram. Switch three (SW3) must be placed horizontally.

Relay Logic Diagrams Rules (5)

- **Rule 7.** Current in the relay logic diagram must flow from left to right

. .

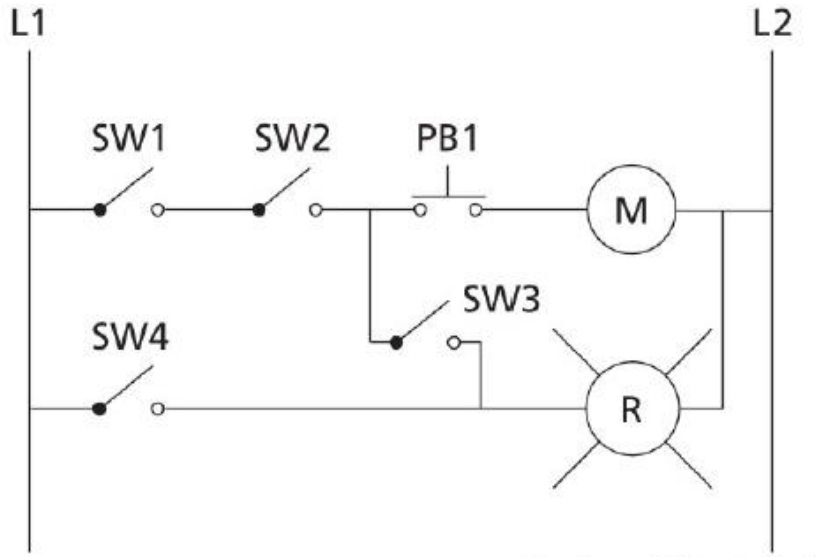


Figure 5-14. A correct relay logic diagram. Input and output devices are placed horizontally.

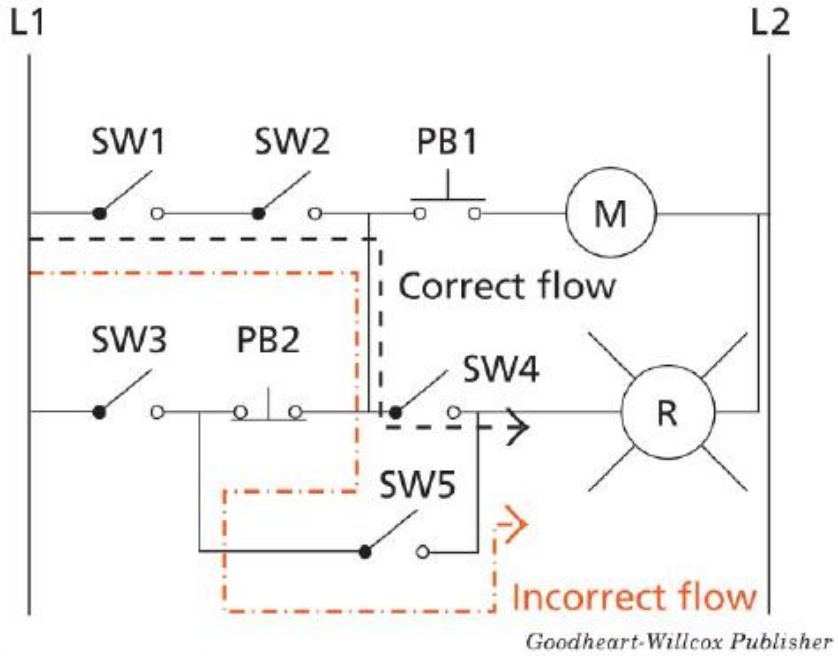
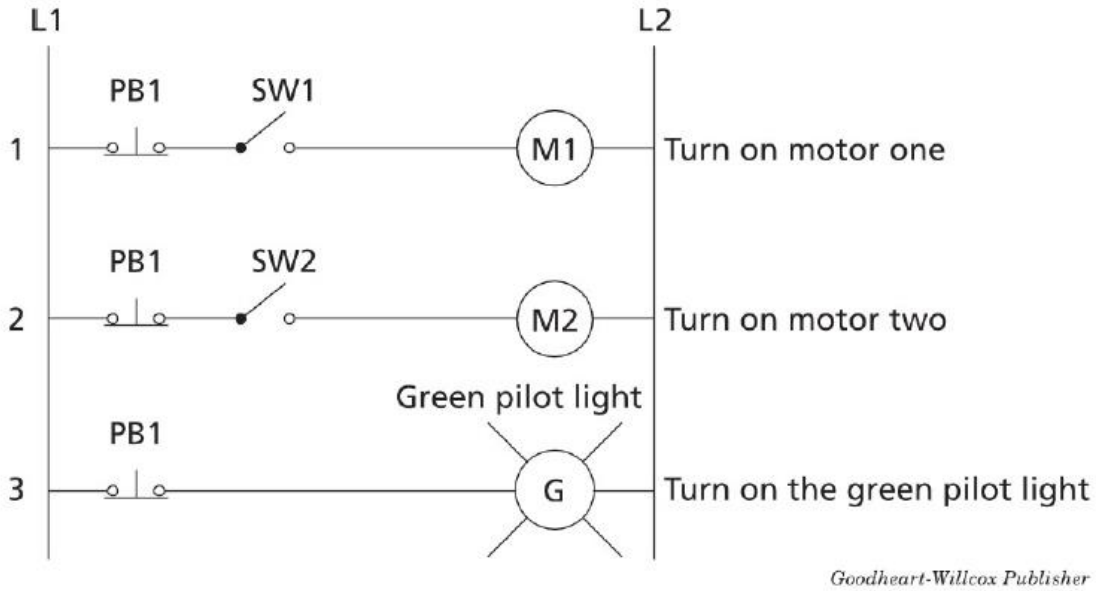


Figure 5-15. Flow process must be from left to right.

Relay Logic Diagrams Rules (6)

- **Rule 8.** Diagrams must be numbered and commented correctly:
  - Rung numbers are placed on left side of each rung
  - Instruction symbols and comments are placed above the instruction
  - Rung comments are placed on the right side of each rung



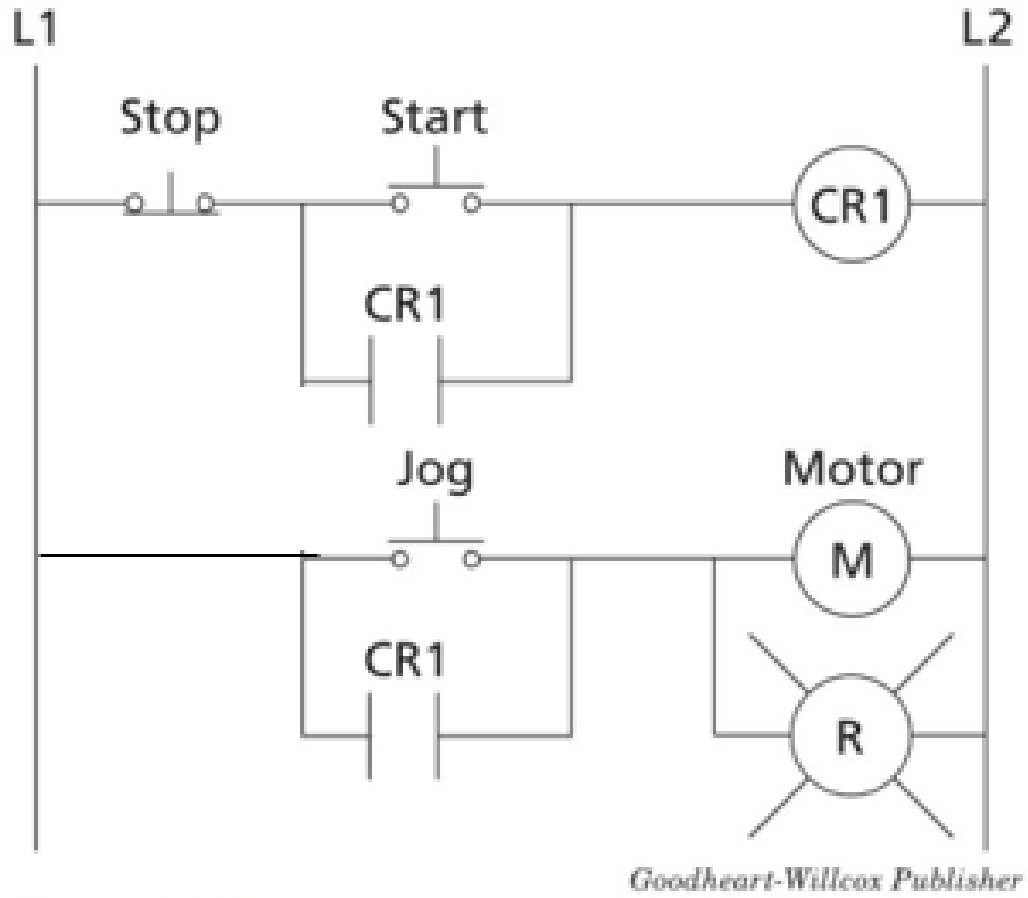
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Figure 5-16. Placing rung numbers by the left rail and comments and contact location by the right rail.

RLD From Text - Example 4 (1)

- A motor can operate continuously when a start NO pushbutton is pressed and released. The same motor will stop its continuous operation when the stop NC pushbutton is pressed and released.
- The same motor is allowed to operate in jog mode. In the jog mode, the motor will operate as long as the NO jog pushbutton remains pressed.
- The red pilot light is on whenever the motor is operating

RLD From Text - Example 4 (2)



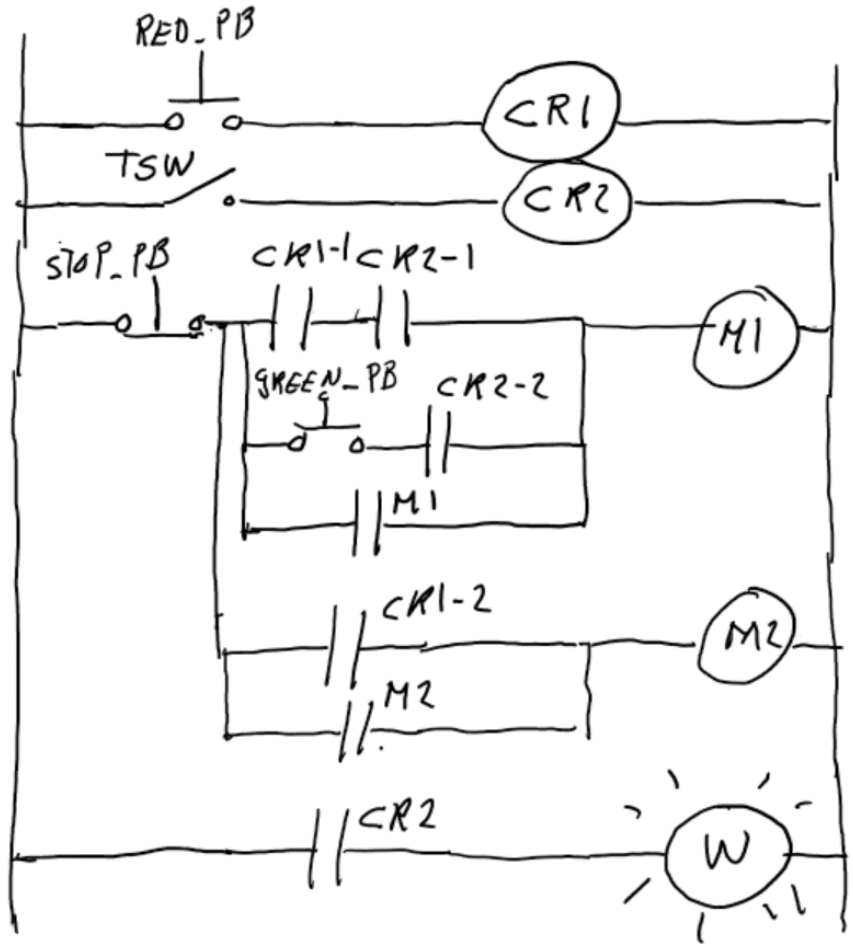
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Figure 5-18. Press the start pushbutton to run the motor continuously. Press the jog pushbutton to run the motor temporarily.

RLD From Text - Example 5 (1)

In a motor control system, one master stop pushbutton (Stop_PB) is available for stopping the operation at any time. If the temperature switch (TSW) is closed, pressing the red pushbutton (Red_PB) will turn on both motors one (M1) and two (M2). If TSW is open, pressing Red_PB will turn on only M2. If TSW is closed and the green pushbutton (Green_PB) is pressed once, M1 will run. Closing the temperature switch turns on the white pilot light.

RLD From Text - Example 5 (2)



- The diagram includes the needed relays to use the RED_PB and TSW more than once.
- Notice that the diagram has only one STOP button (since it is a master button to turn motors off).
- The stop button to shut off the white lamp makes no sense as this button will have to be held down to keep the lamp off. It’s not practical.